

Ahaaram Multi Cuisine Restaurant

Project Report

B.Tech Information Technology

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Abstract

Ahaaram Multi Cuisine Restaurant is a full-stack web application developed to provide an online platform for restaurant services. Customers can explore a multi-cuisine menu, view food details, search and filter items, manage a cart, complete checkout, place orders, reserve tables, manage profiles, view histories, save favourites and submit reviews. An admin dashboard supports food, order, reservation, customer and review management. The application uses HTML5, CSS3, JavaScript, Python, Flask and SQLite.

1. Introduction

Restaurant services involve menu management, customer orders, table reservations and feedback. Ahaaram brings these activities together in a single web application, providing separate customer and administration functions.

2. Problem Statement

Manual or separate processes for menu viewing, ordering, reservations and record management can make restaurant operations harder to organize. The project provides a centralized web solution for these activities.

3. Objectives

1. Develop a web-based restaurant application.
2. Display multi-cuisine food items.
3. Provide search and filtering.
4. Provide cart and checkout.
5. Support online food ordering.
6. Support table reservation.
7. Maintain order and reservation history.
8. Provide favourites and reviews.
9. Provide an admin dashboard.
10. Store application data using SQLite.

4. Existing System

In a conventional restaurant process, menu viewing, ordering, reservations and record maintenance may depend on separate manual interactions. This can make tracking and management less convenient.

5. Proposed System

The proposed system provides one web application where customers can browse food, manage a cart, place orders, reserve tables, view activities and submit reviews. Administrators can manage food, orders, reservations, customers and reviews.

6. System Requirements

Software: Python, Flask, SQLite, HTML5, CSS3, JavaScript and a modern web browser.

Hardware: A computer capable of running Python and a web browser.

7. Technologies Used

Frontend: HTML5, CSS3, JavaScript.

Backend: Python and Flask.

Database: SQLite.

Additional: Werkzeug and Requests.

8. System Architecture

The application uses a web-based architecture. The browser provides the interface, Flask handles routes and application logic, and SQLite stores restaurant data. The admin interface uses the same backend for management operations.

9. System Modules

Customer Module – registration, login and profile.

Menu Module – food categories, search, filtering and details.

Cart Module – selected items, quantities and totals.

Order Module – checkout, order placement and history.

Reservation Module – table reservation and history.

Favourite Module – saved food items.

Review Module – customer feedback.

Admin Module – dashboard, food, orders, reservations, customers and reviews.

10. Database Description

SQLite stores restaurant application records such as customers, food items, orders, order items, table reservations, favourite food items and reviews.

11. Implementation

The application is implemented using Flask routes and HTML templates. CSS and JavaScript provide presentation and interaction. SQLite provides persistent storage. Customers use public pages while administrative functions are provided through the admin dashboard.

12. Results / Output

The application provides menu browsing, food details, cart, checkout, order placement, table reservation, profile and history functions. The administrator can manage the main restaurant records through the dashboard.

13. Advantages

- Centralized restaurant management.
- Easy menu browsing.
- Online ordering workflow.
- Online table reservation.
- Customer history management.
- Admin management interface.
- SQLite-based data storage.

14. Limitations

The current system is a web application and requires deployment configuration for production use. Online payment, real-time delivery tracking, advanced analytics and multi-branch management are outside the current scope.

15. Future Enhancements

Online payment integration, real-time order tracking, email/SMS notifications, restaurant analytics, multiple branches, loyalty programs, coupons, AI-based food recommendations, mobile application support and cloud deployment.

16. Conclusion

Ahaaram Multi Cuisine Restaurant demonstrates how a full-stack web application can combine restaurant ordering, table reservation, customer interaction and administration in one platform. The system provides a base for future payment, delivery, analytics and mobile features.

17. References

1. Python and Flask web development concepts.
2. SQLite database concepts.
3. HTML5, CSS3 and JavaScript web development concepts.
4. Ahaaram Multi Cuisine Restaurant project implementation.